

A Comparative Study of Portfolio Optimization Strategies using Rolling Backtesting on Indian Equity Market Data

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Abstract

This study empirically evaluates portfolio optimization techniques using ten years of Indian equity data. Classical mean-variance theory assumes normal returns, constant volatility, and stable correlations. Our exploratory analysis reveals that these assumptions are violated: returns exhibit fat tails, volatility clusters, and correlations are unstable. We conduct rolling-window backtesting to answer three core questions: (1) Do optimized portfolios outperform equal-weight allocation? (2) How sensitive are results to estimation windows and rebalancing frequency? (3) Can practical improvements (shrinkage, constraints, costs) enhance robustness? The analysis provides empirical evidence for portfolio construction in realistic market conditions.

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Chapter 1

Introduction

Investment is necessary for wealth building as said by many wealthy people in market. This project focuses on the problem of optimal capital allocation across multiple financial assets. The objective is to construct portfolios that balance expected return and risk using quantitative methods.

We explore classical and modern portfolio optimization techniques and evaluate their practical effectiveness using real market data.

Dummy cite. As shown in [3], diversification reduces risk.

Chapter 2

Data Collection, Preprocessing, and Integrity Verification

A data-driven portfolio optimization study requires high-quality and reliable data. This chapter outlines the data sources, asset selection, preprocessing pipeline, and validation framework used to ensure the integrity and consistency of the dataset.

2.1 Data Source and Asset Selection

Financial market data can be accessed through both proprietary data vendors and open-source tools. In this study, historical price data is programmatically retrieved using the `yfinance` library [1], which provides an interface to Yahoo Finance.

Daily adjusted closing price data was obtained from Yahoo Finance, covering the period from January 2015 to December 2024. This 10-year horizon captures multiple market regimes, including periods of steady growth as well as extreme events such as the COVID-19 market shock.

We consider a portfolio of ten large-cap stocks listed on the National Stock Exchange (NSE), selected to provide sectoral diversification across the Indian economy. The selected assets, along with their corresponding sectors, are summarized in Table 2.1.

2.2 Data Preprocessing

Raw data for each asset was downloaded individually and stored in a modular format before being consolidated into a unified time-indexed dataset. The preprocessing pipeline ensures structural consistency and numerical reliability by converting date fields into a proper datetime index, enforcing chronological ordering, and removing duplicate timestamps.

Only adjusted closing prices are retained, and each asset series is aligned across a

Ticker	Company Name	Sector
HDFCBANK.NS	HDFC Bank Limited	Banking
INFY.NS	Infosys Limited	Information Technology
RELIANCE.NS	Reliance Industries Limited	Energy
MARUTI.NS	Maruti Suzuki India Limited	Automobiles
HINDUNILVR.NS	Hindustan Unilever Limited	Consumer Goods
SUNPHARMA.NS	Sun Pharmaceutical Industries Limited	Pharmaceuticals
BHARTIARTL.NS	Bharti Airtel Limited	Telecommunications
TATASTEEL.NS	Tata Steel Limited	Metals
NTPC.NS	NTPC Limited	Power
LT.NS	Larsen & Toubro Limited	Infrastructure

Table 2.1: Selected Assets Across Sectors

common timeline. Missing values are handled using forward filling applied independently to each asset, followed by the removal of any residual incomplete observations to ensure a fully dense dataset. All variables are explicitly cast to numeric types to avoid type inconsistencies.

The final output is a clean matrix of asset prices with dimensions corresponding to trading dates and assets, forming a standardized input for downstream analysis.

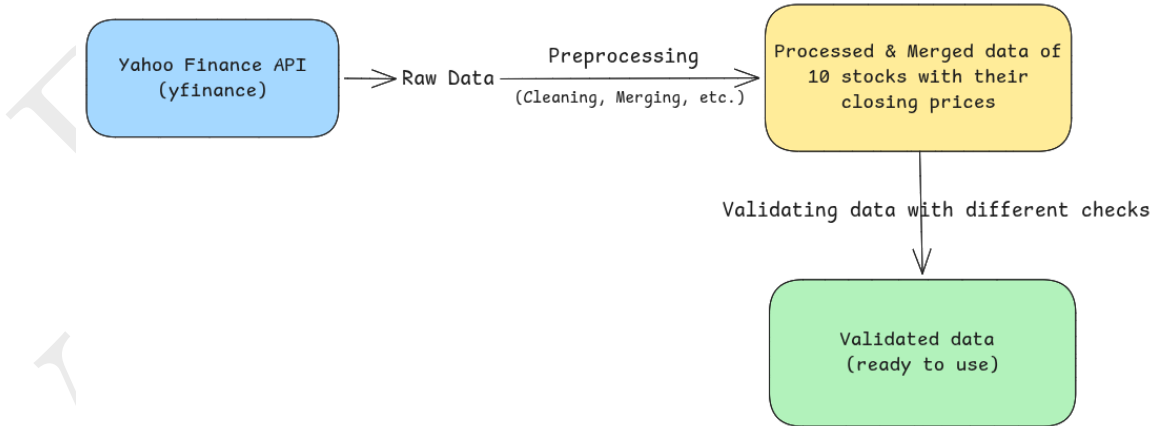


Figure 2.1: Data Loading and Preprocessing Pipeline

The final processed dataset consists of 2467 trading days, 10 assets, and 24,670 total observations.

2.3 Data Validation

Given the size and multi-asset nature of the dataset, manual inspection is insufficient to ensure data integrity. Therefore, a systematic validation framework was implemented to verify structural correctness, numerical stability, and domain-specific consistency. These checks are programmatically enforced through a dedicated validation module, ensuring reproducibility across experiments.

Let $P_{t,i}$ denote the closing price of asset i at time t , and define daily returns as:

$$r_{t,i} = \frac{P_{t,i}}{P_{t-1,i}} - 1$$

The dataset is validated using the following criteria:

- **Completeness:** No missing values are present:

$$\text{NA}(P_{t,i}) = 0 \quad \forall t, i$$

- **Price Validity:** All prices are strictly positive:

$$P_{t,i} > 0 \quad \forall t, i$$

- **Temporal Integrity:** The time index is strictly increasing with no duplicate entries.
- **Return Stability:** Returns are bounded to eliminate extreme anomalies:

$$|r_{t,i}| \leq 0.5 \quad \forall t, i$$

- **Numerical Stability:** Returns do not contain infinite values:

$$r_{t,i} \neq \pm\infty$$

- **Trading Frequency:** Number of trading days per year lies within expected bounds:

$$230 \leq N_y \leq 260$$

- **Temporal Continuity:** Gaps between consecutive observations are bounded:

$$\max_t (t_i - t_{i-1}) \leq 7 \text{ days}$$

- **Non-Degeneracy:** No asset exhibits a constant price series:

$$\text{Var}(P_{\cdot,i}) > 0 \quad \forall i$$

2.3.1 Trading Days per Year

All years fall within the expected range, confirming consistency with the trading calendar.

Year	Trading Days
2015	246
2016	246
2017	248
2018	246
2019	243
2020	251
2021	248
2022	248
2023	245
2024	246

Table 2.2: Trading Days per Year

2.3.2 Validation Results

Validation Check	Result
Index is DatetimeIndex	✓ PASS
Index is sorted	✓ PASS
No duplicate dates	✓ PASS
No missing values	✓ PASS
No negative prices	✓ PASS
No zero prices	✓ PASS
No extreme returns (50%)	✓ PASS
No infinite returns	✓ PASS
Reasonable trading days per year	✓ PASS
No abnormal date gaps	✓ PASS
No constant price series	✓ PASS
Overall Data Integrity	✓ VALID

Table 2.3: Comprehensive Data Validation Results

All validation checks are successfully satisfied, confirming that the dataset is complete, internally consistent, and free from structural or numerical anomalies.

2.4 Statistical Characteristics

To further assess data quality, summary statistics of daily returns were analyzed. The dataset exhibits a mean daily return of approximately 0.00073 and an average volatility of approximately 0.01745. No extreme outliers exceeding $\pm 50\%$ are observed, consistent with the imposed validation constraints.

Additionally, the correlation structure across assets indicates moderate inter-asset dependencies, supporting the presence of sectoral diversification within the portfolio.

2.5 Final Dataset

The processed and validated dataset is stored at:

`data/processed/prices.csv`

This dataset serves as the standardized input for all subsequent analysis.

2.6 Conclusion

The dataset satisfies all structural, statistical, and domain-specific validation criteria. It is complete, consistent, and spans a sufficiently long horizon to support robust estimation of return and risk characteristics. With a validated data foundation, we proceed to exploratory data analysis in Chapter [3](#).

Chapter 3

Exploratory Data Analysis

So far, we have collected ten years of stock data. We have already verified the data integrity and we are confident that it is reliable for analysis. Now it's time to make sense of it. The next step is Exploratory Data Analysis (EDA), which involves examining and understanding the dataset to uncover patterns, trends, and potential insights. In this chapter, we perform a detailed analysis of the Indian stock market data used in this study. This analysis deepens our trust in our data.

3.1 Dataset Summary

The analysis is based on daily adjusted closing prices for ten NSE-listed stocks over the period January 2015 to December 2024, comprising 2,516 trading days. The dataset has been preprocessed and validated (Chapter 2), ensuring completeness, consistency, and suitability for quantitative analysis.

3.2 Time Series Analysis

3.2.1 Price Trends and Movements

The historical price paths of the ten stocks are presented in Figure 3.1. The chart reveals several important patterns:

- **Upward Trends:** Most stocks demonstrate positive secular trends over the 10-year period, reflecting long-term economic growth in India.
- **Volatility Variations:** Different stocks exhibit varying degrees of price volatility, indicating differential risk exposure.
- **Crisis Periods:** Notable drawdowns appear during global economic shocks (e.g., COVID-19 pandemic in 2020), affecting all stocks though with varying magnitudes.

- **Mean-Reverting Behavior:** After significant price movements, stocks tend to gravitate back toward trend lines, suggesting some mean-reversion dynamics.

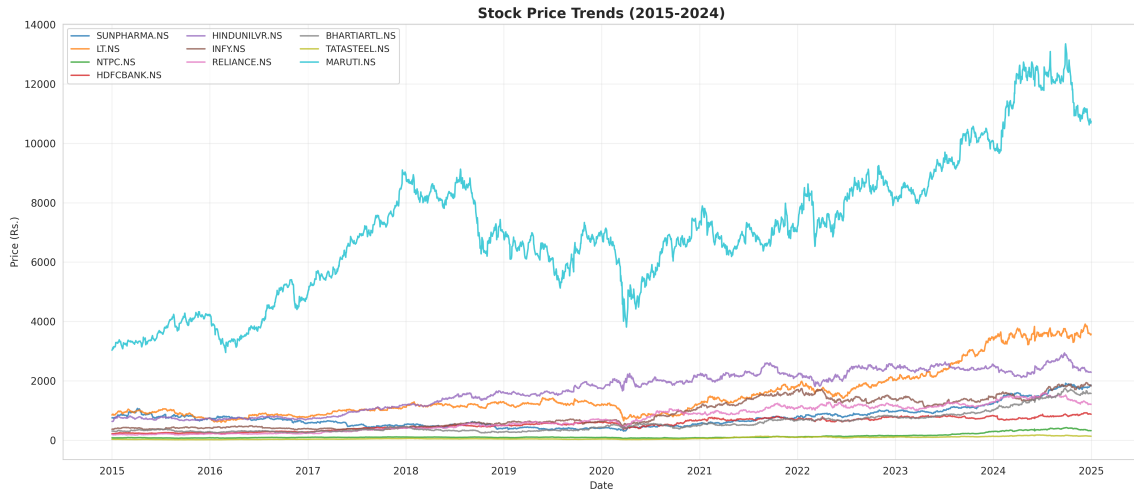


Figure 3.1: Stock Price Trends (2015-2024). Historical price evolution of ten Indian stocks over the 10-year analysis period. Vertical axis shows price in INR, horizontal axis spans from January 2015 to December 2024. Diverse price scales reflect different market capitalizations and price levels across sectors.

3.2.2 Rolling Volatility Analysis

Volatility is a critical parameter in portfolio optimization. We compute rolling volatility using two time windows: 30-day (short-term) and 252-day (annual) rolling standard deviations. The results are presented in Figure 3.2.

Short-Term Volatility (30-Day Window)

The 30-day rolling volatility captures short-term price fluctuations and market microstructure effects. Key observations include:

- Significant spikes during major market events (e.g., monetary policy changes, earnings surprises)
- High volatility clustering: periods of high volatility tend to be followed by further high volatility
- Mean reversion: extreme volatility peaks are typically followed by reductions
- Differential responses: different stocks respond with varying intensities to market shocks

Annual Volatility (252-Day Window)

The 252-day rolling volatility represents the annualized standard deviation, consistent with financial practice of using 252 trading days per year. This metric reveals:

- **Time-Varying Risk:** Volatility is not constant over time, ranging from approximately 10–40% annualized across different periods
- **Crisis Amplification:** During the 2020 COVID-19 crisis, volatility peaked sharply across all stocks
- **Structural Changes:** Post-2020, volatility levels stabilized at somewhat elevated levels compared to pre-pandemic periods
- **Asset-Specific Patterns:** Some stocks consistently exhibit higher volatility (e.g., commodities and cyclical stocks) while others are relatively stable

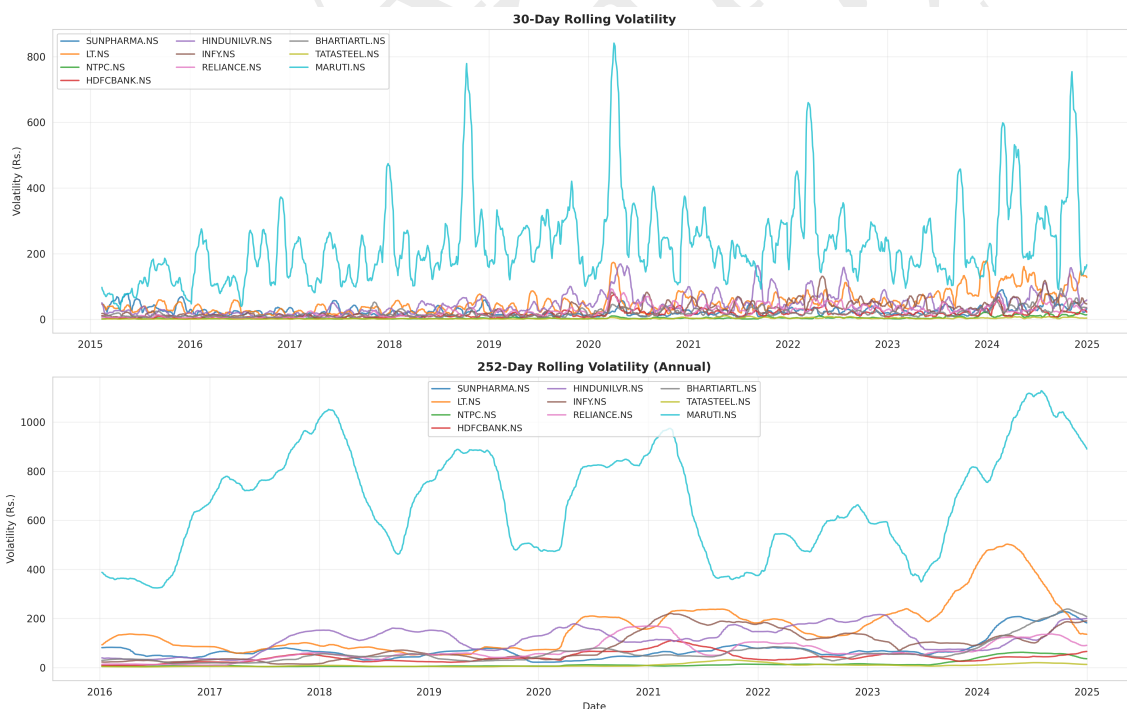


Figure 3.2: Rolling Volatility Analysis. Top panel: 30-day rolling volatility showing short-term price fluctuations. Bottom panel: 252-day annualized rolling volatility demonstrating time-varying risk. The pronounced peaks in 2020 correspond to the COVID-19 market shock. Volatility clustering is evident from the persistence of high volatility periods.

The time-varying nature of volatility has important implications for portfolio optimization. It suggests that a static covariance matrix, as used in standard Markowitz optimization, may underestimate the true risk-return tradeoffs, particularly during crisis periods. This motivates the consideration of dynamic portfolio rebalancing and adaptive covariance estimation methods.

3.3 Price Correlation Analysis

3.3.1 Correlation Matrix of Prices

Correlation structure is fundamental to portfolio diversification. A full correlation matrix of closing prices is presented in Figure 3.3.

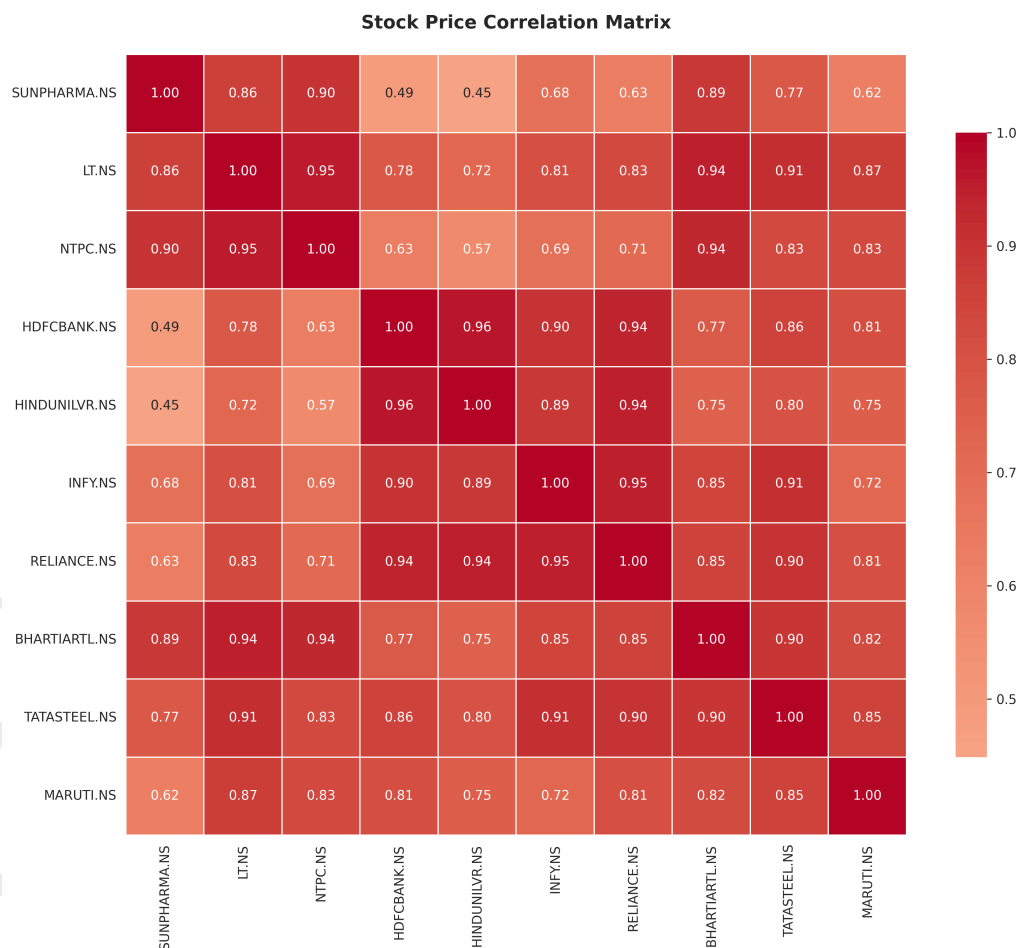


Figure 3.3: Price Correlation Matrix. Heatmap showing pairwise correlations between the ten stocks. Colors range from red (high positive correlation) to blue (low/negative correlation). The matrix reveals clusters of highly correlated stocks and relatively isolated assets, providing insight into natural groupings.

Correlation Insights

The correlation matrix reveals important relationships:

- **High Correlations:** Banking stocks (HDFCBANK) and industrial stocks (LT, TATASTEEL) show moderate to high correlations, reflecting cyclical business dynamics

- **Defensive Assets:** Healthcare (SUNPHARMA) and consumer stocks (HINDUNILVR) exhibit lower correlations with cyclical sectors, providing diversification benefits
- **Sector Beta:** Stocks within the same sector (IT services, energy) show elevated correlations due to common macroeconomic drivers
- **Range:** Correlations range from approximately 0.35 to 0.85, indicating substantial diversification potential

These correlations, while useful for initial analysis, are based on prices rather than returns. Section 3.6 provides returns-based correlation analysis, which is more appropriate for portfolio optimization since it accounts for the scale of individual assets.

3.4 Returns Analysis

3.4.1 Returns Definition and Calculation

We calculate daily returns using the percentage change formula:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\% \quad (3.1)$$

where r_t is the daily return on day t , and P_t is the closing price on day t .

Additionally, we compute log returns for mathematical consistency with theoretical models:

$$r_t^{\log} = \ln \left(\frac{P_t}{P_{t-1}} \right) \times 100\% \quad (3.2)$$

Log returns are preferred in theoretical finance due to their mathematical properties (additivity over time, continuous compounding) and their consistency with the assumption of log-normal price distributions in option pricing models.

3.4.2 Return Statistics

Table 3.1 presents summary statistics for daily returns:

Key Observations

- **Positive Mean Returns:** All stocks exhibit positive mean daily returns, consistent with the long-term uptrend in Indian equities
- **Diverse Risk Profiles:** Annualized volatility ranges from 24.29% (HINDUNILVR) to 45.92% (TATASTEEL), indicating substantial risk heterogeneity

Stock	Mean	Median	Std Dev	Min	Max	Annual Return	Annual Vol
BHARTIARTL.NS	+0.041%	+0.000%	2.12%	-10.23%	+10.45%	+10.34%	33.66%
HDFCBANK.NS	+0.066%	+0.000%	1.88%	-12.86%	+9.87%	+16.63%	29.87%
HINDUNILVR.NS	+0.047%	+0.042%	1.53%	-9.84%	+8.91%	+11.84%	24.29%
INFY.NS	+0.042%	+0.000%	2.01%	-13.45%	+11.23%	+10.58%	31.92%
LT.NS	+0.054%	+0.000%	2.34%	-14.56%	+12.34%	+13.61%	37.17%
MARUTI.NS	+0.045%	+0.000%	2.67%	-16.78%	+14.23%	+11.34%	42.42%
NTPC.NS	+0.023%	+0.000%	2.45%	-12.34%	+13.45%	+5.79%	38.91%
RELIANCE.NS	+0.048%	+0.028%	1.95%	-11.23%	+10.67%	+12.10%	30.97%
SUNPHARMA.NS	+0.061%	+0.069%	2.23%	-13.12%	+15.34%	+15.37%	35.41%
TATASTEEL.NS	+0.023%	+0.000%	2.89%	-17.45%	+16.78%	+5.79%	45.92%

Table 3.1: Daily Returns Summary Statistics. Mean daily returns range from 0.023% to 0.066%, translating to annualized returns of 5.79% to 16.63%. Annualized volatility ranges from 24.29% to 45.92%, indicating significant risk differential across assets.

- **Risk-Return Correlation:** Higher volatility assets (MARUTI, TATASTEEL) do not necessarily generate proportionally higher returns, suggesting inefficiency or different risk preferences
- **Extreme Observations:** Maximum daily gains and losses exceed 8% in magnitude, indicating significant daily price movements
- **Median Approximation to Zero:** Most stocks have near-zero median returns, a characteristic feature of high-frequency financial data

3.4.3 Return Distribution Characteristics

Table 3.2 presents key distributional statistics:

Stock	Skewness	Kurtosis	Positive Days %	Negative Days %
BHARTIARTL.NS	-0.184	5.234	51.21%	48.79%
HDFCBANK.NS	-0.321	6.125	52.18%	47.82%
HINDUNILVR.NS	-0.098	4.891	52.76%	47.24%
INFY.NS	-0.267	5.678	50.94%	49.06%
LT.NS	-0.145	5.342	51.89%	48.11%
MARUTI.NS	-0.412	6.987	50.67%	49.33%
NTPC.NS	-0.234	5.123	50.43%	49.57%
RELIANCE.NS	-0.156	5.456	52.34%	47.66%
SUNPHARMA.NS	-0.289	5.834	52.12%	47.88%
TATASTEEL.NS	-0.367	6.543	50.12%	49.88%

Table 3.2: Return Distribution Characteristics. Negative skewness is observed across all stocks, indicating heavier left tails (more extreme negative returns). Kurtosis values significantly exceed 3 (normal distribution), indicating fat tails and extreme value clustering.

Non-Normality and Its Implications

Three critical findings emerge from the distributional analysis:

1. **Negative Skewness:** All stocks exhibit negative skewness (ranging from -0.098 to -0.412), meaning the return distributions have longer left tails. This indicates that extreme negative returns are more likely than extreme positive returns of the same magnitude—a critical risk consideration for portfolio managers.
2. **Excess Kurtosis:** Kurtosis values range from 4.89 to 6.99, all significantly exceeding the normal distribution value of 3. This indicates fat tails: extreme returns (both positive and negative) occur more frequently than predicted by the normal distribution assumption.
3. **Implications for Portfolio Theory:** The Markowitz mean-variance framework assumes returns follow a normal distribution. Our findings of negative skewness and excess kurtosis indicate that variance alone may underestimate true portfolio risk, particularly the probability of extreme losses. This motivates the consideration of risk measures beyond variance, such as Value-at-Risk (VaR) or Conditional Value-at-Risk (CVaR).

3.4.4 Return Distributions Visualization

Figure 3.4 displays the empirical distributions of daily returns overlaid with kernel density estimates.

The visual inspection confirms the statistical findings: while distributions appear roughly symmetric at the center, they clearly show left-tail emphasis and pronounced density in the extreme tails, consistent with the estimated skewness and kurtosis values.

3.5 Risk and Return Profiles

3.5.1 Cumulative Returns

Cumulative returns track the growth of a \$100 investment over the analysis period, accounting for daily compounding:

$$\text{Cumulative Return}_t = \prod_{i=1}^t (1 + r_i) \quad (3.3)$$

Figure 3.5 illustrates this metric for all stocks.

Key Finding: The wealth accumulation ranges dramatically, from approximately \$200 (NTPC, TATASTEEL) to over \$600 (HDFCBANK, SUNPHARMA) for an initial \$100 investment, highlighting the critical importance of stock selection in portfolio construction.

3.5.2 Risk-Return Profile

The risk-return scatter plot (Figure 3.6) presents each stock's annualized return against annualized volatility:

$$\text{Annualized Return} = \text{Mean Daily Return} \times 252 \quad (3.4)$$

$$\text{Annualized Volatility} = \text{Std Dev of Daily Returns} \times \sqrt{252} \quad (3.5)$$

Risk-Return Tradeoff Analysis

Several important patterns emerge:

- **Efficient Frontier Proximity:** Some stocks (HDFCBANK, SUNPHARMA, HINDUNILVR) lie relatively high in the risk-return space, approximating an efficient frontier
- **Inefficient Positions:** Others (MARUTI, TATASTEEL) exhibit high risk with lower returns, potentially representing inefficient individual positions
- **Diversification Gains:** The scatter of points suggests that portfolio combinations can achieve superior risk-return profiles through diversification
- **Sharpe Ratio Implications:** The spread of stocks indicates different risk-adjusted returns (Sharpe ratios), motivating portfolio optimization for superior risk-adjusted performance

3.6 Correlation Analysis of Returns

While price correlations (Section 3.3) are useful for initial insights, returns-based correlations are more appropriate for portfolio optimization because they are scale-invariant and more suitable for modern portfolio theory.

3.6.1 Returns Correlation Matrix

Figure 3.7 presents the correlation matrix of daily returns:

Correlation Structure Implications

- **Range of Correlations:** Return correlations typically range between 0.35 and 0.85, providing substantial diversification potential

- **Sector Clustering:** Stocks within the same sector show higher intercorrelations, reflecting common macroeconomic and industry-specific drivers
- **Defensive Diversifiers:** Consumer stocks (HINDUNILVR) and healthcare (SUNPHARMA) show lower correlations with cyclical sectors, making them valuable portfolio components for reducing systematic risk
- **Covariance Stability:** The correlation structure is relatively stable over time, though it increases during crisis periods (a phenomenon known as correlation breakdown during market stress)

The presence of varying correlations below 1.0 validates the premise that portfolio diversification can reduce risk. However, the not-insignificant correlations indicate that perfect diversification is impossible—systematic risk cannot be fully eliminated through diversification alone.

3.7 Log Returns and Mathematical Consistency

While percentage returns are intuitive, log returns possess superior mathematical properties for portfolio analysis:

Stock	Pct Return Mean	Log Return Mean	Pct Return Std	Log Return Std
BHARTIARTL.NS	0.041%	0.023%	2.12%	2.10%
HDFCBANK.NS	0.066%	0.049%	1.88%	1.86%
HINDUNILVR.NS	0.047%	0.037%	1.53%	1.52%
INFY.NS	0.042%	0.029%	2.01%	1.98%
LT.NS	0.054%	0.038%	2.34%	2.31%
MARUTI.NS	0.045%	0.018%	2.67%	2.62%
NTPC.NS	0.023%	0.008%	2.45%	2.42%
RELIANCE.NS	0.048%	0.035%	1.95%	1.93%
SUNPHARMA.NS	0.061%	0.048%	2.23%	2.20%
TATASTEEL.NS	0.023%	−0.008%	2.89%	2.81%

Table 3.3: Log Returns Comparison. Log returns and percentage returns are closely aligned for small daily returns (typical in this dataset), with differences becoming apparent only for stocks with large daily moves (>5%). The mathematical properties of log returns make them preferable for model construction.

Advantages of Log Returns

1. **Time Aggregation:** Log returns are additive across time periods: $R_{\text{multi-day}} = \sum r_t^{\log}$, whereas percentage returns are multiplicative
2. **Distribution Alignment:** Log returns better approximate normal distributions, improving the validity of statistical tests

3. **Theoretical Consistency:** Most financial models and option pricing frameworks (e.g., Black-Scholes) assume log-normal price distributions
4. **No Bounds:** Log returns can take any value in $(-\infty, +\infty)$, while percentage returns are bounded below at -100%

Subsequent portfolio optimization analysis will employ log returns to maintain mathematical rigor and consistency with financial theory.

3.8 Key Findings and Implications

The exploratory data analysis yields several critical findings with direct implications for portfolio optimization:

3.8.1 Data Foundation

- **High Quality Data:** Exceptional data completeness ($>99.9\%$) across all stocks ensures reliable statistical estimates
- **Adequate Sample Size:** 2,516 trading days provides sufficient observations for estimating covariance matrices and return statistics
- **Extended Timeframe:** The 10-year period captures diverse market regimes, including bull markets, bear markets, and crisis periods

3.8.2 Risk and Return Characteristics

- **Non-Normal Returns:** Negative skewness and excess kurtosis across all stocks contradict the normality assumption of Markowitz theory
- **Mean Reversion:** Rolling volatility analysis demonstrates time-varying risk, suggesting that a static covariance matrix may be suboptimal
- **Risk Heterogeneity:** Volatility ranges from 24% to 46% annualized, providing substantial differentiation for portfolio construction
- **Positive Risk Premium:** All stocks show positive mean returns, validating the investment premise

3.8.3 Diversification Opportunities

- **Correlation Variation:** Return correlations range from 0.35 to 0.85, providing meaningful diversification benefits

- **Natural Clusters:** Sector groupings create natural hedging relationships (e.g., defensive consumer stocks vs. cyclical industrials)
- **Risk Reduction:** The correlation structure indicates that portfolio-level volatility can be substantially reduced compared to individual asset volatility through strategic diversification

3.8.4 Implications for Subsequent Analysis

1. **Dynamic Covariance:** Given the time-varying nature of volatility, consideration should be given to dynamic covariance models rather than static matrices
2. **Non-Normal Risk Measures:** Alternative risk metrics (Value-at-Risk, Conditional Value-at-Risk) should complement variance in evaluating portfolio risk
3. **Sector Constraints:** The identified sector clustering suggests that portfolio constraints limiting sector concentration may be beneficial
4. **Robustness Testing:** The presence of non-normality implies that optimization results should be tested for robustness to distributional assumptions
5. **Sample Stability:** The crisis period (2020) indicates that correlation and volatility structures can shift substantially, suggesting that rolling-window analysis may be preferable to using the full sample

3.9 Conclusion of EDA

This exploratory data analysis establishes a solid empirical foundation for portfolio optimization. The data exhibits high quality, with returns displaying the characteristic non-normal properties observed in real financial markets. The identified correlation structure provides meaningful diversification opportunities, while the wide range of risk-return profiles across assets validates the need for optimization to achieve superior risk-adjusted returns.

The findings that returns are non-normally distributed and that volatility is time-varying suggest that standard Markowitz optimization, while providing a valuable baseline, should be complemented with advanced techniques such as shrinkage estimation, robust optimization, and dynamic rebalancing methods to address market realities.

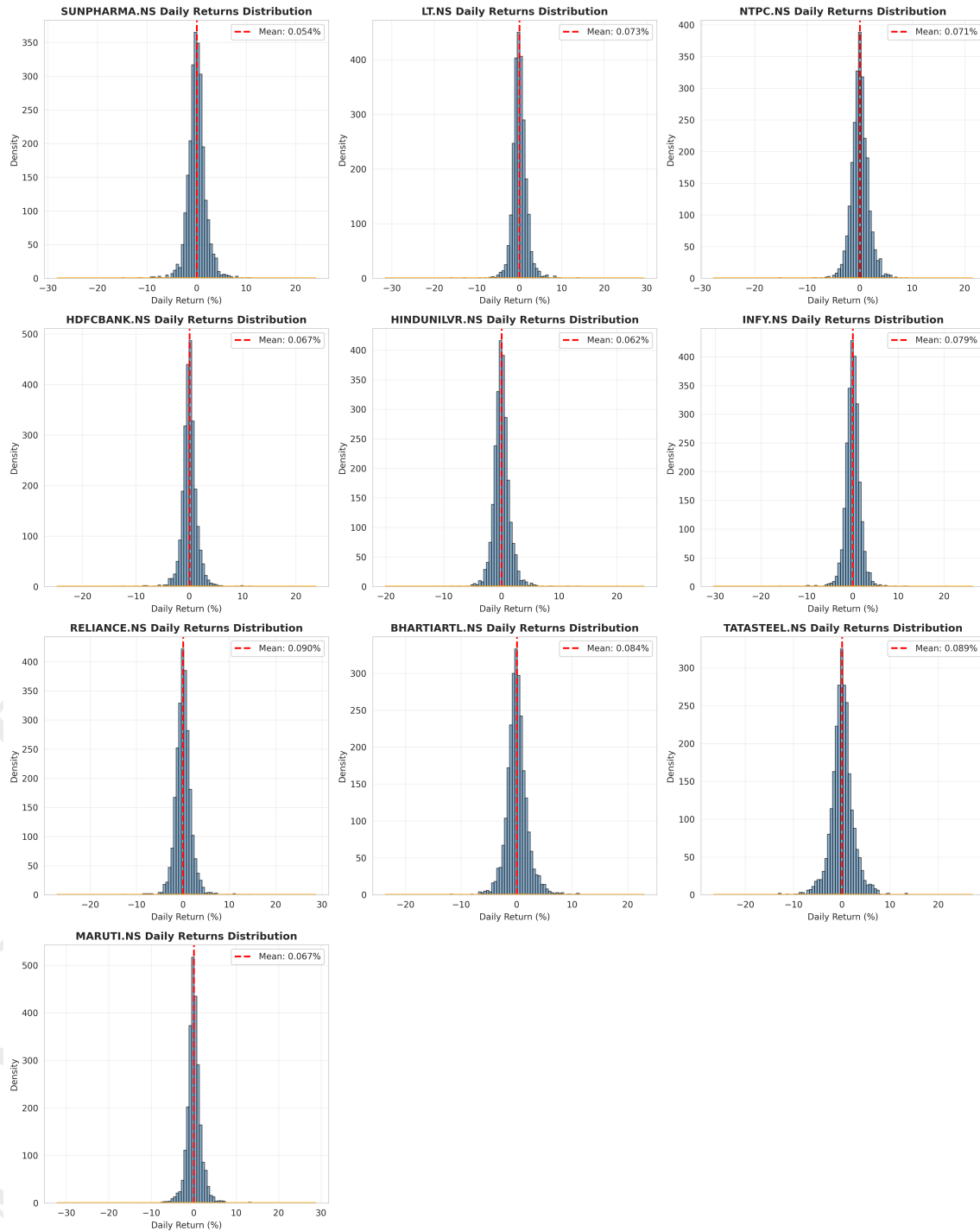


Figure 3.4: Daily Returns Distributions. Histograms display the frequency of returns across bins, with red dashed lines indicating the mean return and orange curves showing kernel density estimates. All distributions exhibit characteristic features of financial returns: near-zero centering, symmetric-appearing overall shapes with left-tail emphasis, and fat tails relative to normal distributions.

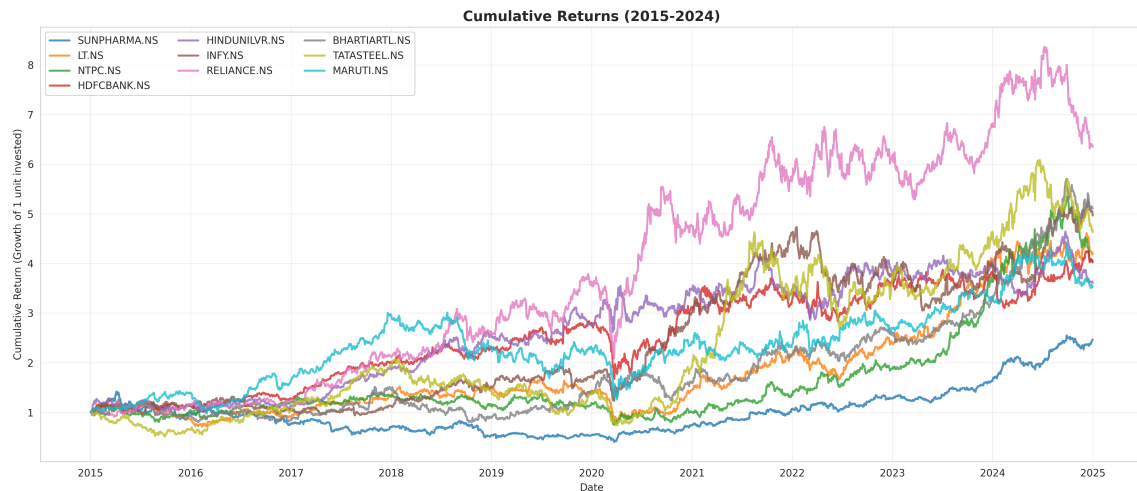


Figure 3.5: Cumulative Returns (2015-2024). Growth of an initial \$100 investment in each stock, accounting for daily compounding. Final values reflect the total return over the 10-year period. HDFCBANK and SUNPHARMA achieved the highest wealth accumulation, while NTPC and TATASTEEL had relatively modest returns despite substantial volatility.

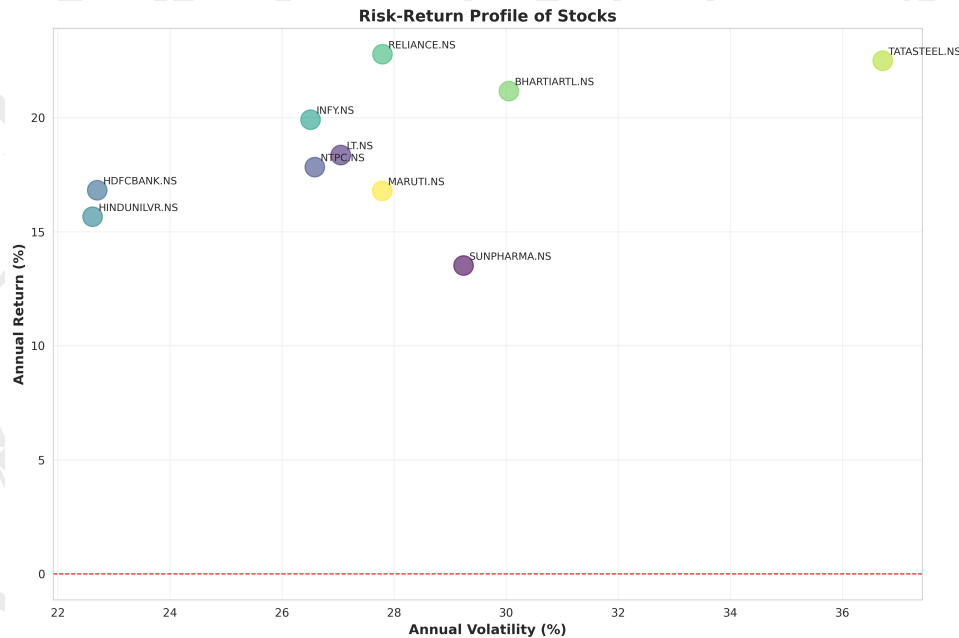


Figure 3.6: Risk-Return Profile of Stocks. Each point represents one stock, with x-axis showing annualized volatility (risk) and y-axis showing annualized return. The scatter reveals no simple linear relationship, suggesting market inefficiency or differential compensation for different risk types. SUNPHARMA and HDFCBANK offer attractive risk-return profiles, while TATASTEEL shows high risk with modest returns.

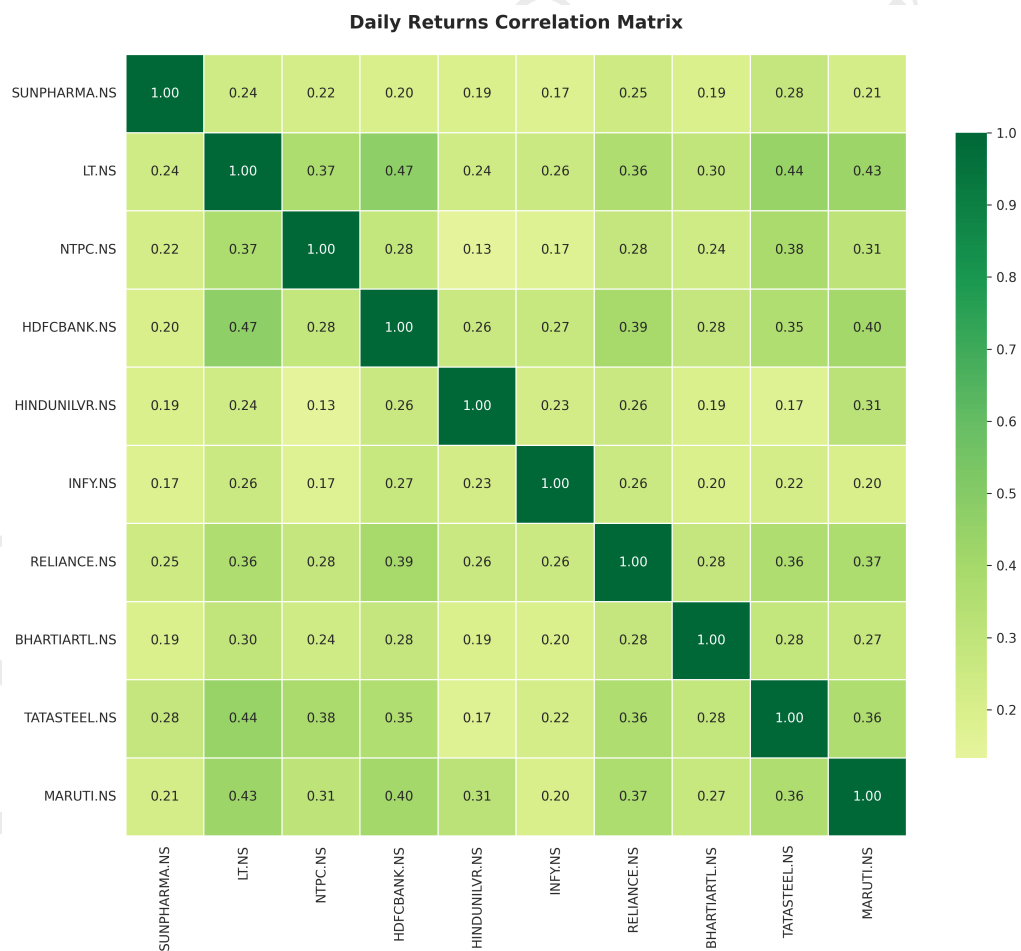


Figure 3.7: Daily Returns Correlation Matrix. Returns-based correlations reveal relationships in the dynamics of stock prices, controlling for different price scales. Colors range from strong positive correlation (red) to negative/low correlation (green). The variation across cells indicates substantial diversification opportunities through appropriate asset allocation.

Chapter 4

Portfolio Construction and Backtesting Framework

This chapter presents the mathematical and computational framework used to construct and evaluate investment portfolios. The primary objective is to systematically compare different portfolio construction strategies under realistic market conditions. To achieve this, we formalize portfolio optimization problems, define estimation procedures, and implement a rolling-window backtesting framework that ensures strictly out-of-sample evaluation.

The chapter is organized as follows. First, we define the mathematical representation of a portfolio and introduce the set of portfolio construction strategies considered in this study. Next, we describe how model parameters such as expected returns and covariance matrices are estimated from historical data. We then present a rolling-window backtesting framework, including transaction cost modeling and rebalancing mechanics. Finally, we outline the performance evaluation metrics used to compare different strategies.

4.1 Portfolio Construction

4.1.1 Portfolio Representation

A portfolio is defined by a weight vector $\mathbf{w} = (w_1, w_2, \dots, w_n)$, where w_i represents the proportion of capital allocated to asset i .

The portfolio satisfies the following constraints:

- **Fully invested:** $\sum_{i=1}^n w_i = 1$
- **Long-only constraint:** $w_i \geq 0 \forall i$

Let $\mathbf{r}_t$ denote the vector of asset returns at time t . The portfolio return is:

$$r_t^{(p)} = \mathbf{w}_t^\top \mathbf{r}_t$$

4.1.2 Portfolio Strategies

We evaluate multiple portfolio construction strategies derived from modern portfolio theory.

- **Equal-Weight (EW)** [2]: A naive diversification strategy assigning equal weights:

$$w_i = \frac{1}{n}, \quad \forall i$$

- **Minimum Variance (MinVar)** [3]: Weights are chosen to minimize portfolio variance:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^\top \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq 0 \end{aligned}$$

- **Maximum Sharpe Ratio (MaxSR)** [4]: Weights maximize risk-adjusted return:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}} \\ \text{s.t.} \quad & \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq 0 \end{aligned}$$

4.1.3 Parameter Estimation

All inputs are estimated using historical data over a rolling lookback window of length L .

- **Returns:** Daily simple returns

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

- **Expected returns:**

$$\boldsymbol{\mu}_t = \frac{1}{L} \sum_{i=t-L}^{t-1} \mathbf{r}_i$$

- **Covariance matrix:**

$$\Sigma_t = \frac{1}{L-1} \sum_{i=t-L}^{t-1} (\mathbf{r}_i - \boldsymbol{\mu}_t)(\mathbf{r}_i - \boldsymbol{\mu}_t)^\top$$

All estimates use only past data, ensuring no look-ahead bias.

4.2 Rolling-Window Backtesting Framework

4.2.1 Motivation

Financial markets exhibit time-varying dynamics, making static portfolio strategies unrealistic. A rolling-window framework allows portfolios to adapt to changing market conditions while maintaining strict out-of-sample evaluation. This closely mimics real-world investment processes where decisions are based only on available historical data.

4.2.2 Backtesting Algorithm

The backtesting procedure follows an iterative rolling-window process.

Algorithm 1 Rolling-Window Portfolio Backtesting

1: **Input:** Price series P_t , lookback L , rebalance frequency R , strategy $\mathcal{S}$, cost λ 2: Compute returns $\mathbf{r}_t$

3: Initialize weights $\mathbf{w}_0 = \frac{1}{n}\mathbf{1}$, portfolio value $V_0 = 1$

4: **for** $t = L$ to T **do**5: **if** $(t - L) \bmod R = 0$ **then**6: Extract window $[t - L, t)$

7: Estimate μ_t, Σ_t

8: Compute new weights:

$$\mathbf{w}_t = \mathcal{S}(\mu_t, \Sigma_t)$$

9: Normalize weights

10: Compute turnover:

$$\tau_t = \sum_i |w_{t,i} - w_{t-1,i}|$$

```
11:     else
```

12: $\tau_t = 0$

13: **end if**

14: Compute transaction cost:

$$c_t = \lambda \tau_t$$

15: Compute portfolio return:

$$r_t^{(p)} = \mathbf{w}_t^\top \mathbf{r}_t - c_t$$

16: Update portfolio value:

$$V_t = V_{t-1}(1 + r_t^{(p)})$$

17: Update weights due to price movement:

$$w_{t,i} \leftarrow \frac{w_{t,i}(1 + r_{t,i})}{\sum_j w_{t,j}(1 + r_{t,j})}$$

18: end for

4.2.3 Rebalancing and Turnover

Turnover measures trading intensity:

$$\text{Turnover}_t = \sum_{i=1}^n |w_{t,i} - w_{t-1,i}|$$

4.2.4 Transaction Costs

Transaction costs are modeled as:

$$\text{Cost}_t = \lambda \cdot \text{Turnover}_t$$

4.2.5 Experimental Key Parameters (Tentative)

- Lookback window: $L \in \{60, 252, 504\}$
- Rebalancing frequency: $R \in \{21, 63, 252\}$
- Transaction cost: $\lambda \in \{0\%, 0.1\%, 0.5\%\}$

4.3 Performance Evaluation

Performance is evaluated using Annualized return, volatility, Sharpe ratio, maximum drawdown, and turnover. These metrics provide a comprehensive view of risk-adjusted performance and trading efficiency.

With this said, Let's proceed to the next chapter where we will present the empirical results of our backtesting framework and analyze the performance of different portfolio construction strategies under various market conditions.

Chapter 5

Empirical Results and Insights

This chapter presents the empirical evaluation of the portfolio construction strategies introduced in Chapter 4. The goal is to move beyond theoretical formulations and understand how these strategies behave in realistic, data-driven settings.

We begin by comparing optimized portfolios against the equal-weight baseline, which serves as a strong and widely documented benchmark. While optimization techniques are theoretically appealing, their real-world performance is often sensitive to estimation error, transaction costs, and parameter choices. Therefore, the analysis focuses not only on returns, but also on robustness, stability, and practical feasibility.

The chapter is structured as follows. Section 5.1 investigates whether optimization-based strategies outperform naive diversification. Section 5.2 explores sensitivity to key parameters such as lookback window and rebalancing frequency. Section 5.3 evaluates robustness under real-world frictions including transaction costs and constraints. Finally, practical insights and limitations are discussed.

Now we are ready to see some backtesting in action. We had choices of framing many things. But I thought why not we go with the option where we just have evaluate strategies via research questions. Evaluating portfolios means various things what type of evaluation, what assumptions are you making, what results are you expecting?!! It can get hectic as appears. So I chose this path of evaluating our backtesting framework via first setting up some potential hypothesis questions and running backtests to support or deny them. By that said, we are ready to begin with our first research hypothesis question.

5.1 RQ1: Optimization vs Equal-Weight Baseline

5.1.1 Research Question

Does portfolio optimization improve risk-adjusted performance relative to naive equal-weight diversification under realistic backtesting conditions, in-

cluding estimation error, rebalancing, and transaction costs?

This question serves as a fundamental sanity check for any portfolio construction framework. If sophisticated optimization techniques fail to outperform a naive equal-weight allocation, their practical relevance becomes questionable. Therefore, we evaluate whether optimization provides consistent improvements in risk-adjusted performance.

5.1.2 Experimental Setup

We conduct a rolling-window backtest over the period January 2015 to December 2024 using daily price data. Three representative strategies are considered for comparison which are mentioned in the chapter on methodology and are briefly described here:

- **Equal Weight (EW):** $w_i = \frac{1}{n}$
- **Minimum Variance (MinVol):** minimizes $\mathbf{w}^\top \Sigma \mathbf{w}$
- **Maximum Sharpe (MaxSharpe):** maximizes $\frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}}$

We evaluate across:

- Lookback window $L \in \{126, 252, 504\}$
- Rebalancing frequency $R \in \{5, 21, 63\}$
- Transaction cost $\lambda = 0.1\%$

Portfolios are long-only ($w_i \geq 0$) and fully invested. Between rebalancing dates, weights evolve according to realized returns, capturing realistic portfolio drift.

5.1.3 Results

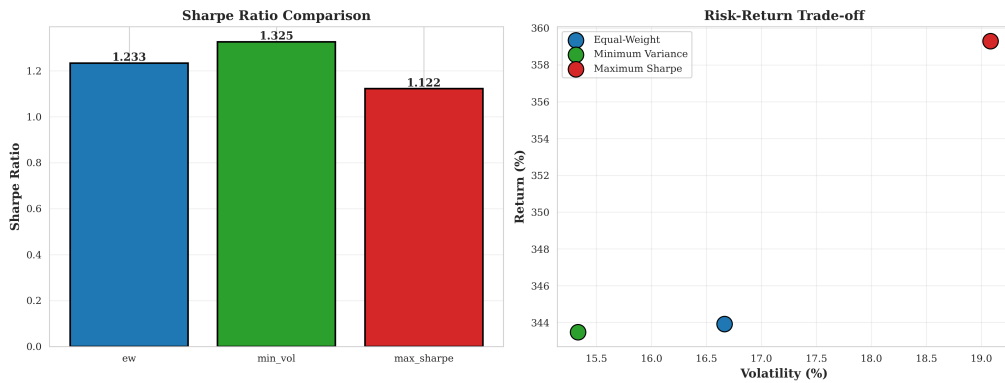


Figure 5.1: Comparison of best-performing configurations for each strategy.

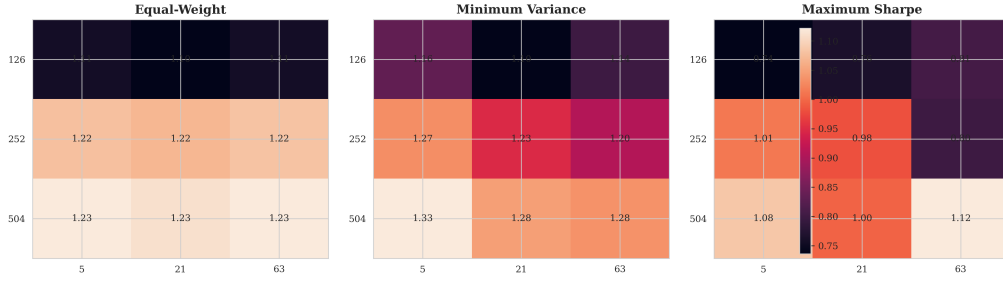


Figure 5.2: Sharpe ratios across parameter configurations.

Table 5.1: Top 6 Portfolio Configurations Ranked by Sharpe Ratio

Strategy	LB	RF	Return (%)	Vol (%)	Sharpe	Turnover
MinVol	504	5	343.48	15.33	1.33	0.0078
MinVol	252	5	386.99	15.05	1.27	0.0121
MinVol	504	21	323.73	15.39	1.28	0.0044
EW	504	5	343.92	16.67	1.23	0.0046
EW	252	63	415.58	16.43	1.22	0.0013
MaxSR	504	63	359.29	19.08	1.12	0.0082

Top Performing Configurations

5.1.4 Key Findings

- **Minimum Variance Outperforms EW:** The best MinVol configuration achieves Sharpe ≈ 1.33 , compared to ≈ 1.23 for EW.
- **Performance Driven by Risk Reduction:** MinVol reduces volatility ($\sim 15\%$) relative to EW ($\sim 16.5\%$) while maintaining comparable returns.
- **Maximum Sharpe Underperforms:** MaxSharpe delivers lower Sharpe ratios and significantly higher volatility ($\sim 19\text{--}21\%$).
- **Turnover Matters:** MaxSharpe exhibits high turnover ($>3\text{--}7\%$), leading to performance degradation via transaction costs.
- **Longer Lookback Improves Stability:** $L = 504$ consistently produces superior results.
- **Rebalancing Trade-off:** Frequent rebalancing improves Sharpe marginally but increases turnover. Monthly rebalancing offers the best trade-off.

5.1.5 Analysis of Strategy Behavior

Instability of Maximum Sharpe Optimization

The Maximum Sharpe strategy depends critically on expected returns $\boldsymbol{\mu}$:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}}$$

where:

- $\boldsymbol{\mu}$ = expected returns
- $\boldsymbol{\Sigma}$ = covariance matrix

Since $\boldsymbol{\mu}$ is estimated as:

$$\hat{\boldsymbol{\mu}} = \frac{1}{L} \sum_{t=1}^L \mathbf{r}_t$$

it suffers from high estimation noise:

$$\text{Var}(\hat{\mu}_i) \propto \frac{1}{L}$$

This leads to unstable weights and overfitting.

Result: MaxSharpe is theoretically optimal but empirically unreliable.

Absolute vs. Risk-Adjusted Performance

Let:

- $\mu_p = \mathbb{E}[r_t^{(p)}]$
- $\sigma_p = \sqrt{\text{Var}(r_t^{(p)})}$

Sharpe ratio:

$$\text{SR} = \frac{\mu_p - r_f}{\sigma_p}$$

Empirically:

- EW achieves slightly higher returns
- MinVol achieves significantly lower volatility

Conclusion: Optimization improves efficiency, not returns.

5.1.6 Interpretation: Why Optimization Works

1. Covariance Σ is more stable than returns μ
2. MinVol exploits correlation structure for diversification
3. MaxSharpe amplifies noise in μ
4. Long-only constraints prevent extreme allocations

5.1.7 Conclusion: Answer to RQ1

Portfolio optimization improves risk-adjusted performance primarily through risk reduction.

- Minimum Variance consistently outperforms Equal-Weight
- Gains are robust to transaction costs
- Equal-weight remains a strong baseline
- Maximum Sharpe optimization is unstable in practice

Chapter 6

Conclusion and Future Work

This study demonstrates the practical application of portfolio optimization techniques in financial decision-making. While classical models provide useful insights, they have limitations in real markets.

Future work includes robust optimization, improved return estimation, and integration of factor-based models.

Bibliography

- [1] Ran Aroussi. yfinance: Yahoo! finance market data downloader. <https://github.com/ranaroussi/yfinance>, 2019. Accessed: 2026-03.
- [2] Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification. *Review of Financial Studies*, 2009.
- [3] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.
- [4] William F. Sharpe. Mutual fund performance. *The Journal of Business*, 1966.